

Subarta Ghosh

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PROFESSIONAL SUMMARY

Full Stack Developer skilled in building scalable web applications using Java, Spring Boot, FastAPI, React, REST APIs, and SQL. Experienced in developing backend systems, implementing caching and asynchronous processing, and integrating AI-powered features into modern applications. Strong foundation in Data Structures & Algorithms, OOPs, DBMS, and full-stack development.

EDUCATION

Master of Computer Applications (MCA) – CGPA: 9.17/10 Vellore Institute of Technology, Vellore	2025–2027
B.Sc. (Hons) Computer Science University of Calcutta, Kolkata	2022–2025
Senior Secondary (CBSE - PCM)	2021–2022
Secondary (ICSE)	2019–2020

TECHNICAL SKILLS

- **Programming Languages:** Java, SQL, Python, JavaScript, C
- **Frontend Technologies:** React.js, HTML5, CSS3, JavaScript
- **Backend & API Development:** Spring Boot 3, REST APIs, Node.js, Express.js, FastAPI
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Platforms:** Git, Docker, GitHub Actions, Render, Vercel
- **Distributed Systems & Caching:** Apache Kafka, Redis, Caffeine, Event-Driven Architecture
- **Monitoring & Performance:** Prometheus, Grafana, Micrometer, Resilience4j, Bucket4j
- **AI & NLP:** RAG, FAISS, NLP Pipelines, TF-IDF, NMF
- **Core CS Concepts:** Data Structures & Algorithms, OOPs, DBMS, Operating Systems

PROJECTS

Asphalt Prep – Adaptive Learning SaaS Platform <i>Spring Boot 3, PostgreSQL, Redis, Caffeine</i> <i>Aiven Kafka, Resilience4j, Bucket4j, Docker</i>	Aug 2025 – Jan 2026 Live Demo: asphaltprep.subartaghosh.co.in Link: github.com/subh-ghosh/ai-practice-platform
• Developed a scalable full stack learning platform using Spring Boot and React with modular backend architecture. • Implemented multi-level caching (Redis + Caffeine), reducing mean latency from 196.6ms to 76.4ms (61.2% faster) and P95 latency from 614.9ms to 116.4ms (81.1% improvement) across 22 endpoints; improved throughput from 14.23 to 15.70 req/s. • Applied Bucket4j token-bucket rate limiting; during stress testing blocked 14,687 requests with a 96.39% block rate. • Integrated Resilience4j patterns (circuit breaker, retry, fallback), reducing stress-path latency from 2226ms to 77ms. • Introduced asynchronous event processing with Kafka, reducing HTTP failure rate from 1.46% to 0.28%. • Integrated LLM-based features using Grok API to generate adaptive learning content and dynamic responses, enhancing personalization within the platform. • Instrumented backend using Actuator, Micrometer, Prometheus, and Grafana for real-time monitoring. • Deployed backend on Render and frontend on Vercel using Dockerized environments.	
Autonomous Fire Detection and Response System <i>Embedded Systems, C++, IoT</i>	Feb 2025 – Jun 2025 Link: github.com/subh-ghosh/arduino-based-automated-fire-fighting-robot
• Built an ESP32/Arduino-based fire detection robot using multi-flame sensors and servo-controlled suppression. • Applied Gaussian-based probabilistic classification replacing static threshold detection. • Modeled stopping distance using Linear Regression and integrated Blynk IoT alert monitoring.	
SignalShift – AI-Powered Customer Feedback Analytics Platform <i>Python, scikit-learn, FastAPI, React, NLP, OpenAI API, FAISS</i>	Feb 2026 – Present Link: github.com/subh-ghosh/signalshift
• Built a scalable backend system using FastAPI to process and analyze 191,000+ Google Play Store reviews, enabling API-driven data pipelines and structured processing workflows. • Implemented a machine learning pipeline using Logistic Regression and TF-IDF Vectorization (Bigrams), achieving 87.6% accuracy and a 0.914 F1-score on test data. • Integrated LLM-based summarization and insight extraction to generate concise summaries and actionable insights from large-scale unstructured text data. • Designed a retrieval-augmented (RAG-ready) architecture using text chunking and embedding-based preprocessing to support semantic search and contextual querying. • Experimented with vector similarity search using FAISS for efficient retrieval of relevant feedback segments. • Applied NMF Topic Modeling to categorize feedback into 12 recurring product issue categories, enabling prioritization of product improvements. • Built REST APIs to expose analytics and AI-powered insights to frontend applications. • Ensured model robustness using 5-fold cross-validation, achieving a low performance variance of 0.0013 across folds.	

ACHIEVEMENTS & CERTIFICATIONS

- Secured Global Rank 840 in **TCS CodeVita Season 13**
 - Certified for **SQL (Advanced)** – HackerRank
 - Solved 150+ Data Structures & Algorithms problems on coding platforms.
- Certificate Link
Certificate Link